

Methods in Cognitive Neuroscience

Mathematical Methods

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Introduction

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Pioneers of Theoretical Biology

D'Arcy Thompson (1920s), Nicolas Rashevsky (1930s) and Alan Turing (1940s) were pioneers in the proposing of a theoretical framework for biology.



Mathematics and Neuroscience

- Schrödinger wrote in **1944** *'What is life?'*, trying to use the mathematics developed by that time to explain biological processes.
- In **1949** psychologist Donald Hebb created a hypothesis of learning based on the mechanism of neural plasticity that is now known as Hebbian learning and its later variants were early models for long term potentiation.
- In **1952** biophysical models of neuron, the Hodgkin-Huxley model was proposed, where each component of an excitable cell was modeled as an electrical element.

History Mapping

"The next stage in the development of biological science will be revolutionary in its conceptual foundations and strongly mathematical in its methods."

-Denis Noble (Oxford biologist)

Component investigated

Mathematical Model

Ion channels, Synapse

Dynamical systems theory

Single nerve cell

Nonlinear stochastic dynamics

Multi unit activity

Graded responses

Columnar activity

Dynamical systems, fibre bundles

Brain modules

Physics of phase transitions

Continuum neural field theory

Brain network of modules

Interacting field theories

Global brain

Response theory

History

Hierarchical Nature

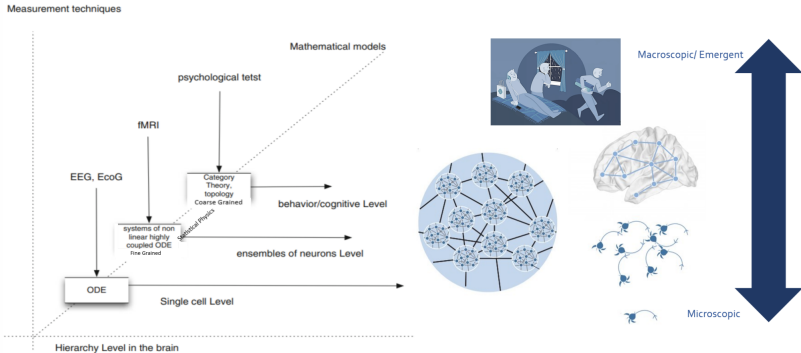


Figure: Measurement techniques, brain hierarchy levels and mathematical models are related as the figure shows.

System and States

We are given the following system:

- A directed graph $\mathbf{G} = (\mathbf{V}, \mathbf{E})$.
- An $\mathbb{R}$ -colouring $\mathbf{h}: \mathbf{V} \rightarrow \mathbb{R}$
- An $\mathbb{R}$ -weighting $\mathbf{w}: \mathbf{E} \rightarrow \mathbb{R}$
- Space of states $\Sigma = \mathbf{V} \rightarrow \{\mathcal{Q}, \mathcal{E}, \mathcal{R}\}$

Here, $\mathbf{V}$ is the set of vertices, $\mathbf{E}$ is the set of directed edges, $\mathbf{h}_i$ is a value assigned to every vertex (the threshold potential), $\mathbf{w}_{ij}$ is the weight attached to each edge (strength of connection).

- $\mathcal{Q}$ - Quiescent
- $\mathcal{E}$ - Excited
- $\mathcal{R}$ - Refractory

A model for the brain

Dynamics

We define a probabilistic evolution by $T: \Sigma \rightarrow \Sigma$ by

$$\sigma_v = \mathcal{Q} \rightarrow T(\sigma)_v = \mathcal{E} \text{ with probability } \begin{cases} 1 & \text{if } \sum_{e \in \mathbf{A}_{\sigma,v}} w_e > h_v \\ r_1 & \text{otherwise} \end{cases}$$

$$\sigma_v = \mathcal{E} \rightarrow T(\sigma)_v = \mathcal{R} \text{ with probability } 1$$

$$\sigma_v = \mathcal{R} \rightarrow T(\sigma)_v = \mathcal{Q} \text{ with probability } r_2$$

where $\mathbf{A}_{\sigma,v} = \{e \in \text{tar}^{-1}\{v\} \mid \sigma_{\text{src } e} = \mathcal{E}\}$ or the exited neighbours

Order Parameters

- Define activity $\mathbf{a}: \Sigma \rightarrow \mathbb{N}$ by $\mathbf{a}_\sigma = \text{card } \sigma^{-1} \{\mathcal{E}\} =$ the # of neurons in the excited state at a particular stage
- An avalanche $\mathcal{A}$ with minimal activity $\mathbf{a}_0$ is a sequence

$$\mathcal{A} = ((n_i, T^{n_i}(\sigma)), \dots, (n_f, T^{n_f}(\sigma)))$$

- $\mathbf{a}_{T^{n_i}(\sigma)}, \mathbf{a}_{T^{n_f}(\sigma)} \leq \mathbf{a}_0$
 - $\mathbf{a}_{T^n(\sigma)} > \mathbf{a}_0$ for $n_i < n < n_f$.
- Avalanche Size: $\mathbf{S}(\mathcal{A}) = \sup_{n \in \{n_i, \dots, n_f\}} \mathbf{a}_{T^n(\sigma)} =$ the maxima of the activity in a particular avalanche
- Avalanche Time: $\mathbf{T}(\mathcal{A}) = n_f - n_i - 1 =$ the duration of the avalanche

Thermodynamics

For an initial state σ and time evolution length n , define

$$N_{\text{size}}(\sigma, a_0, s, n) = \langle \text{Number of avalanches of minimal activity } a_0 \\ \text{and size } s \text{ in the sequence } (\mathcal{T}^k(\sigma))_{k=0}^n \rangle$$

$$N_{\text{time}}(\sigma, a_0, m, n) = \langle \text{Number of avalanches of minimal activity } a_0 \\ \text{and time } m \text{ in the sequence } (\mathcal{T}^k(\sigma))_{k=0}^n \rangle$$

$$\Gamma(\sigma, a_0, m, n) = \langle \text{Average size of avalanches of minimal activity } a_0 \\ \text{and time } m \text{ in the sequence } (\mathcal{T}^k(\sigma))_{k=0}^n \rangle$$

where the $\langle \dots \rangle$ average is over all possible time evolutions of σ , weighted by the appropriate probabilities.

A model for the brain

Thermodynamics

$$f_{\text{size}}(a_0, s) = \left\langle \lim_{n \rightarrow \infty} \frac{N_{\text{size}}(\sigma, a_0, s, n)}{n} \right\rangle_{\sigma}$$

$$f_{\text{time}}(a_0, m) = \left\langle \lim_{n \rightarrow \infty} \frac{N_{\text{time}}(\sigma, a_0, m, n)}{n} \right\rangle_{\sigma}$$

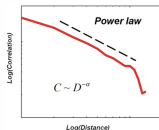
$$\gamma(a_0, m) = \left\langle \lim_{n \rightarrow \infty} \Gamma(\sigma, a_0, m, n) \right\rangle_{\sigma}$$

- $f_{\text{size}}(a_0, s)$ is the probability an avalanche of minimal activity a_0 and size s occurs
- $f_{\text{time}}(a_0, m)$ is the probability an avalanche of minimal activity a_0 and time m occurs
- $\gamma(a_0, m)$ is the average size of an avalanche of minimal activity a_0 that lasts for a time m

Phase Transition and Criticality

The system goes through a phase transition at points where thermodynamic quantities show non-analytic behaviour. They are characterized by:

- **Power Law Behaviour**



- **Scaling Behaviour** The exponents of various thermodynamic quantities are related to each other by simple relations.
- **Scale Invariance** Zooming in or out in the scales retain the behaviour, there is a self-similarity is a characteristic.
- **Universality** Many vastly different systems have the same behaviour at the critical point.

Universal Critical Dynamics in High Resolution Neuronal Avalanche Data

Nir Friedman,¹ Shinya Ito,² Braden A. W. Brinkman,¹ Masanori Shimono,^{2,5} R. E. Lee DeVille,³ Karin A. Dahmen,¹
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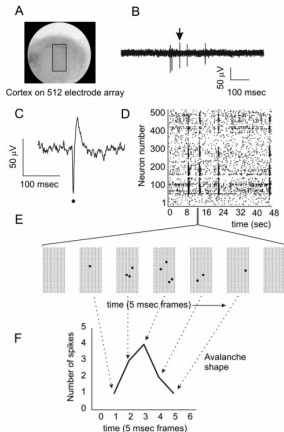
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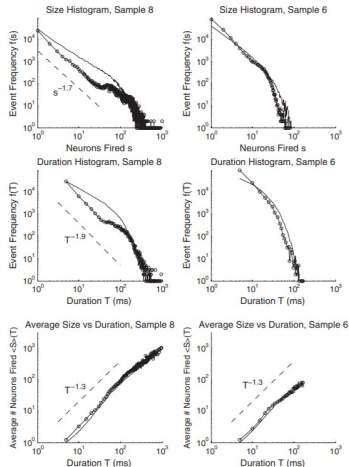
(Received 8 December 2011; revised manuscript received 27 February 2012; published 16 May 2012)

To function optimally, neuronal networks have been hypothesized to operate near a nonequilibrium critical point. The researchers show that the dynamics of cultured cortical networks are critical.



They removed 400 μm thick slices of cortical tissue from living rats, cultured them until they reached maturity (approximately one month), and placed them on an array of 512 electrodes spaced 60 μm .

Experiment used drugs to manipulate network connectivity by altering number of connections and connection strength.



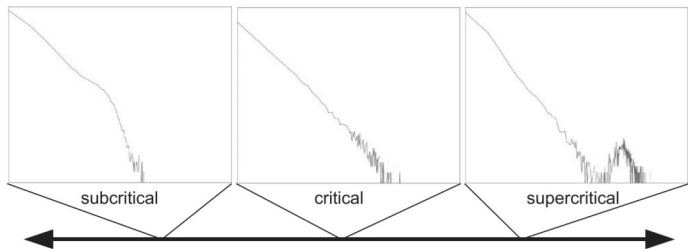
The left graphs (Sample 8) shows the **critical** behaviour, where as the right graphs (Sample 6) shows signatures and deviations from criticality.

- $f_{\text{size}}(0, s) \sim s^{-\alpha}$
- $f_{\text{time}}(0, m) \sim m^{-\tau}$
- $\gamma(0, m) \sim m^{-\frac{1}{\nu}}$

The relationship between the 'critical exponents':

$$\frac{\alpha - 1}{\tau - 1} = \frac{1}{\nu}$$

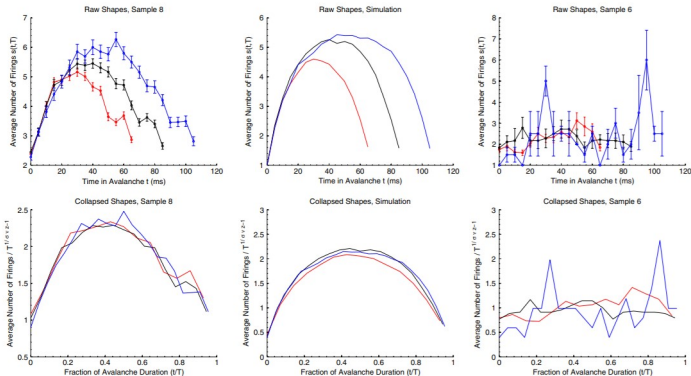
The Transition



Avalanche size histograms from three different samples of rat cortex.

The horizontal line represents a path in some representative (such as manipulating mean connection strength or varying number of connections)

Literature Scaling and Data Collapse



On dividing all avalanche curves by their duration and rescaling their heights by their duration raised to an exponent ν that is related to the critical exponents α and τ , the critical curves collapse on each other.



**Brain Organization into Resting State Networks Emerges at Criticality
on a Model of the Human Connectome**

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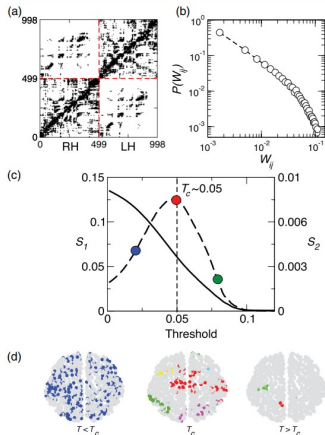
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(Received 24 September 2012; revised manuscript received 1 February 2013; published 22 April 2013)

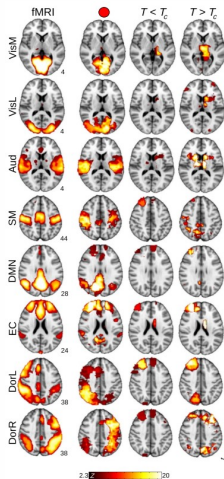
They approach the problem of finding a relation between large-scale brain structure and function by studying the dynamical regime under which realistic spatio-temporal patterns of brain activity emerge from the empirically derived network of human brain neuro-anatomical connections. The results show that **critical dynamics unfolding** on the structural connectivity of the human brain.



The model. (a) shows the model adjacency matrix (i.e., the connectome) and (b) the edge weights distribution.

Data obtained from the white matter tracts connecting a parcellation of 998 nodes covering the entire human brain. (c) shows the giant cluster's size, (i.e., the order parameter, S_1 , solid line) and the second largest cluster's size (S_2 , dashed line) as a function of the threshold h (control parameter) as well as the critical point $h_c = 0.05$. (d) illustrates examples of clusters (denoted with different colors) at the three h values denoted with the filled circles in panel (c)

Literature Results



The spatial clusters of coherent activity, termed resting state networks (RSN) are specifically associated with neuronal systems responsible for sensory, cognitive, and behavioral functions. The spatial organization of the human brain RSN emerges spontaneously in the model near T_c . RSN obtained from the model are shown for three values of $h = h_c$, $h < h_c(0.03)$, $h > h_c(0.1)$, and the correlation maxima with the human RSN fMRI, which is always h_c .

Advantages

- We can put to our use a great deal of methods already developed in physics such as non-equilibrium statistical physics, quantum field theory (CFT), Renormalisation Group theory etc. to understand the dynamical properties of the brain as a whole.
- These methods have well defined algorithms to compute n -point functions which can give a theoretical backing to correlators observed from fMRI studies and further enhance our structure to function understanding of the brain.
- These mathematical ideas give solid proofs for the optimal and efficient functioning of the brain which earlier could only be approached evolutionary reasoning.
- We can hope to understand the behaviour as an emergent property of the brain as a whole.

Limitations

- The models we are choosing are simplistic and hence cannot give a complete understanding of brain dynamics as incorporating all aspects will make the model too complicated to work with (with present methods).
- Communication gap between streams of analytical subjects (mathematics, physics, theoretical computer science) and soft subjects (neuroscience, psychology) which slows down absorption of methods and understanding each others fields.

Perspective

Perspectives

There are two ways of modelling cognitive systems.

- One is meant to demonstrate that we understand some particular aspects of the system, and the other tries to discover new structural and functional relationships in the system.
- The former allows numerical simulation, while the latter attempts to capture the organising principles underlying the behaviour of the system.
- The former is data driven, while the second is driven by the creative act of the modeller and ultimately endorsed by the experimental data.

Remarks

In the end, brain function and behaviour depend upon the **interaction** of elements (cells, networks, brains).

The neural and cognitive sciences, deal with complex components (assemblies of cells, brains) and shall inevitably follow a **mathematical strand**...

References

The items are hyperlinks.

- [Universal Critical Dynamics in High Resolution Neuronal Avalanche Data](#), Nir Friedman et al.
- [Brain Organization into Resting State Networks Emerges at Criticality on a Model of the Human Connectome](#), Haimovici et al.
- [Book: A New Foundation for Representation in Cognitive and Brain Science](#), Gómez Ramirez, Jaime
- [Being critical of criticality in the brain](#), Beggs and Timme
- [Phase Transition Videos](#)